

The Arrow as the Collapse Variable: Event Density and Gravitational State Reduction

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July 2026

Series: Event Density (ED) Generative Papers — measurement / state-reduction positioning note (physics-papers/state-reduction/). It positions ED's existing account of quantum state reduction (the arrow that selects the pointer basis and is gravity's time; ED Report §4 and §6) against the two nearest published neighbors, the gravitational-collapse program of Diósi and Penrose.

Status: draft positioning note, honestly tiered. It introduces no new derivation; it locates a standing ED account relative to named published rivals, marks what ED shares, what it offers each of them, where it diverges, and the one thing it owes them.

anchors (ED side, cited not re-derived): Report §4 (the arrow selects the pointer basis; einselection primitive; the Gleason/Piron-Solèr reconstruction; Born non-contextuality; entanglement as the pre-individuation regime, Paper_072), Report §6 (the unification move: the pointer-basis selector *is* the khronon; the problem of time dissolves; unitarity emergent, not fundamental), GR-II (the khronon), the A1 channel-capacity result (controlled capacity between causally disconnected regions is exactly zero). **External:** L. Diósi, *Phys. Lett. A* **120**, 377 (1987); R. Penrose, *Gen. Rel. Grav.* **28**, 581 (1996).

Preamble — What This Note Does NOT Claim

1. **No new result.** It positions the *existing* ED measurement/unification account (Report §4, §6) against two published collapse models. The ED mechanisms are cited, not re-derived.
2. **ED gives a scaling argument for the collapse rate (companion note); the gap is *narrowed*, not closed.** It reproduces the Penrose-Diósi *scaling* $\tau \propto \hbar/E_G$ from ED's primitives (the khronon-clock difference + einselection supplying Penrose's missing preferred time) — but E_G is identified with Penrose's answer rather than computed, and the coefficient and exact energy functional are open. It further shows the 2020 collapse-radiation bound that excluded parameter-free DP *does not exclude ED*, for a computed reason: ED's Lindblad collapse operators are discrete channel projectors ($\hat{J}_\alpha = \hat{g}_\alpha \Pi_\alpha$, FORCED from P11), so a committed charge is an exact fixed point of the dissipative face and does not continuously diffuse — unlike CSL/DP's position-coupled noise. This resolves what first looked like a tension with P11's line-gap face; the one load-bearing question it rested on — is ED's channel basis coarser than ℓ_{ED} ? — is answered grounded-in-kind (companion StateReduction_ChannelGranularity.md: emergent unitarity makes the grain coherent evolution not measurement; einselection is environment-set and coarse; distinguishability is operational). It is *immunity, not a win*: standard QM predicts the same null observable; ED's distinction is structural (a genuine collapse theory on the

non-radiating side), not a new prediction. §5 states this honestly; it is the note’s center, not a footnote.

3. **The core ED identification is a structural synthesis, not a theorem.** That the einselection arrow and the gravitational khronon are one primitive is argued from the sole-process finding, not forced by a deeper principle, and it carries no delivered prediction yet (Report §6’s own guardrail, carried here verbatim in spirit).
4. **Not a refutation.** This is a positioning: generous where ED shares these programs’ convictions and can offer them something, honest where it diverges, and explicit about the number it owes them.

Abstract

There is a serious minority program that says the collapse of the wavefunction is not observer-magic but a real physical effect, and that its cause is **gravity**. Lajos Diósi (1987) got it first: refuse to quantize the gravitational field, treat it instead as fundamentally indeterminate — a universal noise — and that noise decoheres superpositions of different mass distributions on a time $\tau \approx \hbar/E_G$, where E_G is the gravitational self-energy of the difference between the two mass configurations. Roger Penrose (1996) gave the deepest version: a superposition of two mass distributions is a superposition of two *spacetimes*, and general covariance makes the notion of “which time is flowing” across them ill-defined, so the very time-translation operator of the superposed state is uncertain; that time-uncertainty is an energy uncertainty E_G , and the state decays in $\tau \approx \hbar/E_G$. Event Density belongs in this room, and it makes a claim that is both stronger and different. Where both say *gravity causes collapse*, ED says measurement’s definiteness and gravity’s time are **one primitive wearing two hats** — the arrow, which is the irreversible act of commitment: the same arrow that selects the quantum pointer basis (einselection) is the khronon, gravity’s preferred time. The collapse variable in ED is therefore **time** (the committed arrow), one step upstream of the **energy** E_G that Penrose’s time-ambiguity produces. This has two consequences. ED *supplies the preferred time Penrose finds missing*, resolving the gap his argument opens rather than competing with it. And ED *dissolves the problem of time* instead of adding a collapse term to Schrödinger, obtaining locality from the *forward* arrow and its capacity-zero result rather than from any hidden variable. ED gives a *scaling argument* for the rate from its own primitives — the two branches run on two commitment-clocks ($N \sim \sqrt{b}$, the khronon being the committed time) that the single arrow cannot hold, so einselection individuates them at $\tau \propto \hbar/E_G$ (companion derivation) — reproducing the Diósi–Penrose scale with a deeper origin, with the coefficient and the exact energy functional left open (E_G is identified with Penrose’s answer, not computed). It carries two refinements over its rivals: a *definite* reduction, not Diósi’s mere decoherence to a mixture; and a *native* regulator for the point-mass divergence. The one experimental threat — the spontaneous-radiation bound that already excludes the simplest Diósi–Penrose model — does *not* exclude ED: its collapse operators are discrete channel projectors (FORCED from P11), so a localized charge is a fixed point of the dissipative dynamics and does not continuously radiate, unlike the continuous-position noise the bound constrains. But that is *immunity, not a distinctive win* — standard quantum mechanics predicts the same null; ED’s distinction here is structural, not a new observable. A candidate weapon (ED’s gravity is MOND, and mesoscopic lab masses are sub- a_0 in self-gravity, so the collapse energy would be MOND-enhanced) is closed by ED’s own External Field Effect, which Newtonizes any Earth-based experiment: $ED \approx$ Penrose in the lab. So the gap is **narrowed, not closed**, and ED sits as a coherent, non-excluded member of the gravitational-collapse program — its advance a deeper reason (the arrow supplies Penrose’s missing time), not yet a delivered number.

§1 — The shared conviction

Standard quantum mechanics leaves the collapse of the wavefunction outside its own equations. Unitary evolution is smooth and deterministic; the definite outcome of a measurement is not; and the theory bridges the two only by invoking an “observer” or a “measurement,” which is a placeholder, not a mechanism. A minority of physicists have refused that placeholder and asked what actually makes an outcome definite. A striking fraction of them arrive at the same suspect: **gravity**.

The reasoning is not arbitrary. Gravity is the one interaction that couples to *everything* and grows with mass, so it is the natural candidate for a process that leaves microscopic superpositions alone (we see interference for electrons) but destroys macroscopic ones (we never see a cat in two places). And gravity is where quantum mechanics and general relativity are known to be incompatible, so it is where a *new* effect, beyond textbook QM, is most expected.

Diósi, Penrose, and Event Density all share two convictions that set them apart from the mainstream:

- **Geometry is not an independent field waiting to be quantized.** This is the quiet radical move. Mainstream quantum gravity quantizes the metric as one more field. Diósi refuses — he treats the Newtonian potential as a classical field with a fundamental indeterminacy. Penrose keeps the geometry semiclassical, so that *the mass distribution is the spacetime*. Event Density goes furthest of all: geometry in ED is **emergent**, $g \sim 1/b$, the coarse-grained shadow of the substrate’s participation density, never a primitive field.
- **Collapse is physical, and it is gravity’s doing.** Not an update of an observer’s knowledge; a real event in the world, rooted in gravity or geometry, that would happen in an empty universe with no observers in it.

Those two commitments are the common ground. The rest of this note is where ED, having entered the room agreeing, then says something the others do not.

§2 — ED’s account of state reduction (cited, not re-derived)

ED’s account of measurement predates this comparison and is stated in the Report; here it is only summarized, with its tiers, so the positioning has something concrete to stand on.

The arrow is the collapse (Report §4). ED has exactly one collapse primitive: **commitment (P11), the irreversible act by which the substrate resolves a participation into a definite channel**. This single primitive delivers the structural facts standard QM leaves unexplained. It selects the **pointer basis**: a superposition is never itself a channel, so committing a superposed state resolves it to one channel or another, never to the superposition — the channel basis is the unique basis in which outcomes are definite, and it is picked out by the arrow, not by environmental decoherence. This is **einselection made primitive rather than derived**. The same irreversibility supplies **repeatability**: a committed channel is locked (an irreversible outcome cannot drift), so re-measurement returns it with certainty — irreversibility is not an obstruction to the quantum-logical covering law, it is what supplies it. And because the bandwidth b_K is an intrinsic state property, the **Born probabilities** $P(K) = b_K / \sum b$ are **non-contextual**, consistent with Gleason and Kochen–Specker at once because outcomes are *stochastically committed, not pre-valued*. *Tier: a coherent reconstruction of quantum kinematics via the Gleason/Piron–Solèr chain, not a closed theorem; orthogonality is reduced to operational distinguishability and the complex field is selected, with Solèr lattice rigor the standing residual.*

The same arrow is gravity’s time (Report §6). ED’s gravity is khronometric: it is built on a preferred foliation, and that foliation is the arrow made dynamical — the **khronon**. So the preferred time that quantum mechanics wants (to make outcomes definite *at* a time, distinguishing before from after) and the preferred foliation that ED’s gravity runs on are **not two structures to be reconciled; they are one primitive, coarse-grained into two sectors**. The “problem of time” — QM wants a preferred time, GR’s diffeomorphism invariance forbids one — does not get solved by compromise; it **dissolves**, because ED genuinely has a preferred time (the arrow), and that arrow does both jobs. *Tier: a structural synthesis, not a theorem. The identification is forced by the finding that commitment is the sole process primitive, which is itself a P11-pivoted audit result, not a proof; and the identification carries no delivered prediction yet. Distinctive, structural, unconfirmed.*

Unitarity is emergent, not fundamental (Report §6). The arrow that forbids reversibility forbids erasure: every commitment is a permanent, un-erasable record. Reversible unitary quantum mechanics is the *between-commitments* description; the substrate itself is irreversible. This is why ED preserves black-hole information without a fundamental Hilbert-space unitarity (the record survives; no firewall). It matters here because the gravitational-collapse program also gives up exact unitarity — deviations from Schrödinger evolution are the whole point — and ED gives that up at the foundational level, by design.

Hold three things from this section: **the arrow selects definiteness (einselection); the arrow is gravity’s time (the khronon); and these are one primitive**. They are what ED brings into the comparison.

§3 — The two neighbors, one at a time

3.1 Diósi (1987): gravity as fundamental indeterminacy

Diósi’s move is the one ED most cleanly shares. He *does not quantize* the gravitational field. Instead he posits that the Newtonian potential is a classical (c-number) field carrying a universal, fundamental indeterminacy — a “gravitational white noise,” $\langle \phi(r, t) \phi(r', t') \rangle = \hbar G |r - r'|^{-1} \delta(t - t')$. Feeding that noise into the Schrödinger evolution of a mass distribution yields a master equation with a damping term that destroys interference between states of different mass distribution, on a characteristic time $\tau_d \approx \hbar / [U(x - x') - U(0)] = \hbar / E_G$. Atoms are untouched (E_G negligible); a colloidal grain loses coherence at its own size. (Diósi flags the point-mass divergence — $E_G \rightarrow \infty$ for point masses — the regularization problem that dogs the whole program.)

Shared: the refusal to quantize gravity, and a universal, mass-scaled, physical decoherence. **Divergence:** Diósi’s collapse is a stochastic *field* injected as an addition to quantum mechanics — a noise term bolted onto the Schrödinger equation. ED’s collapse is *internal to the ontology*: it is the arrow, commitment, the same primitive that constitutes time and selects the pointer basis. ED does not add a noise to QM; ED’s QM is the emergent, between-commitments description of a substrate whose fundamental act is the commitment. Diósi supplies a mechanism ED would call a *shadow* — a coarse-grained effective description — of something more primitive.

3.2 Penrose (1996): the ill-defined time, and ED’s answer to it

Penrose’s is the deepest version, and it is the one that touches ED’s nerve directly. Take a lump of material in a quantum superposition of two locations. Each location, on its own, is a stationary state with a well-defined mass distribution and therefore a well-defined spacetime geometry. The superposition

is thus a superposition of **two different spacetimes**. Now ask for the state’s energy, i.e. the generator of time-translation. In general relativity, by diffeomorphism invariance, there is no coordinate-independent way to identify the points of one spacetime with the points of the other — so there is **no well-defined time-translation operator for the superposed state**. That ill-definedness is an irreducible uncertainty in the state’s energy, which in the Newtonian limit equals E_Δ , the gravitational self-energy of the *difference* between the two mass distributions. By the energy–time relation, the superposition is not stationary after all: it decays, like an unstable particle, in a lifetime

$$T \approx \frac{\hbar}{E_\Delta}.$$

Penrose is careful that this is not a theory of collapse, only an argument for the *level* at which gravity must force deviations from unitary quantum mechanics; and he argues (against Diósi) that his version should conserve energy, protected by general covariance.

Here is the ED point, and it is the sharpest paragraph in this note. **Penrose’s mechanism is the ill-definedness of a preferred time**. His superposition collapses *because* there is no fact about which time is flowing across the two superposed spacetimes. Event Density’s entire foundation is that there **is** such a fact: the arrow, the substrate’s direction of commitment, is a primitive preferred time — and (Report §6) it is *the same* preferred time gravity’s khronon foliation picks out. So ED does not stand beside Penrose as a rival; **ED supplies exactly the object whose absence drives his argument**. Read Penrose forward: “there is no well-defined time across the superposition, therefore it must reduce.” Read ED: “there is a well-defined committed time — the arrow — and the commitment in that time is the reduction.” ED locates the collapse in the **time** itself (the committed arrow), one step *upstream* of the **energy** E_Δ that Penrose derives as the time-ambiguity’s consequence. Penrose finds the collapse in the energy uncertainty; ED finds it in the resolution of the very time-ambiguity Penrose starts from.

This is why the resonance is not decorative. The two accounts are about the *same variable* — the preferred time of a superposed gravitational configuration — approached from opposite directions: Penrose from its absence in GR, ED from its presence in the substrate.

A note on locality, which is where ED quietly earns its keep. A deep problem in the whole gravitational-collapse program is *non-locality*: once the gravitational self-energy is large enough to matter, the superposition is already spread across space, and a local collapse would seem to require reaching across it instantaneously. Approaches that keep hidden variables tend to buy locality only by paying in superdeterminism (a teleological, end-state-dependent rule that violates measurement independence). ED does not pay that price. ED’s collapse is a **local substrate event** — each commitment happens at a locus — and it is **forward-causal**: the arrow runs one way, no end-state input. ED’s A1 result establishes that the controlled information capacity between causally disconnected substrate regions is *exactly zero*, so locality is built in, not bought; and ED’s outcomes are **stochastic at the moment of commitment**, not fixed by hidden variables. So ED delivers a local, physical collapse with the *forward arrow* rather than with a superdeterministic hidden variable — a genuine foundational advantage worth stating plainly.

§4 — The unifying difference: time, not energy; one primitive, not a causal chain

Step back from the two and the common structure appears. Diósi and Penrose each say: **gravity** → **collapse**. There is quantum mechanics, and there is a gravitational effect *added* to it (a noise, an energy uncertainty) that causes the collapse, with E_G setting its scale. Collapse is a *consequence* of gravity.

Event Density says something structurally different: **the committed arrow-time is gravity's time and is the pointer-basis selector — one primitive, two hats, not a causal chain.** Definiteness and gravitational time are not cause and effect; they are the same thing seen in two sectors. This is why ED does not *add* a collapse term to the Schrödinger equation: in ED the Schrödinger equation is the emergent, between-commitments description, and the commitment (the collapse) is the more fundamental act, not a correction to it. And it is why ED dissolves the problem of time rather than reconciling it: there is one preferred time, it is the arrow, and it does every job that needs a preferred time — including making outcomes definite and running gravity's foliation.

The collapse *variable* is therefore different. For the program, it is **energy** (E_G , the gravitational self-energy of the difference). For ED, it is **time** (the committed arrow), from which any energy scale would be a derived consequence — which is exactly the direction Penrose's own argument runs (time-ambiguity *first*, energy uncertainty *derived*). ED's claim is that the program has been working one level downstream of the real variable.

§5 — The honest gap: ED owes them a number

Everything above is ED's advantage in *depth of reason*. Here is the program's advantage in *sharpness of result*, and it is not small.

Diósi and Penrose both deliver the same quantitative, parameter-free, falsifiable prediction: $\tau \approx \hbar/E_G$. That is a number, testable in the laboratory. And it *is* being tested. A 2020 underground experiment searched for the spontaneous radiation the collapse must emit and found none at the predicted level, **excluding the simplest, parameter-free (natural-regularization) Diósi-Penrose model.** These models put a number on the table and let the world shoot at it. That is what a physical theory is supposed to do, and it is why the program is taken seriously.

Event Density can offer a *scaling argument* for this number, worked out in the companion note (StateReduction_CollapseRate_ED_Derivation.md). It runs through ED's own primitives: the two branches of the superposition source two bandwidth fields, hence two lapses $N \sim \sqrt{b}$, hence — because the khronon *is* the committed time (§5, and the proper-time form is GR-I) — two *commitment-clocks* running at rates differing by $\Delta\Phi/c^2$. A single arrow cannot hold two branches whose committed-times run at different rates, so einselection individuates them when the accumulated relative commitment-action reaches $\hbar$, giving $\tau \propto \hbar/E_G$. This **reproduces the Diósi-Penrose scaling** — with a deeper origin (the arrow, not E_G , is fundamental; ED supplies the well-defined preferred time whose *absence* drives Penrose's own argument) and two refinements over the rivals: the outcome is a *definite, recorded reduction* (not Diósi's mere decoherence to a mixture — the same einselection that gave the pointer basis in §4); and the point-mass divergence is regulated *natively* by the finite substrate grain and binding-mass extent (where Diósi and Penrose must insert a cutoff by hand). Two honest limits: the quantitative content is still Penrose's (ED's genuine addition is the *preferred time*, not a new number), and the step that names the branch-clock-difference energy E_G *identifies* it with Penrose's self-energy rather than *computing* it from the participation fields — so what is delivered is the scaling and its origin, with the coefficient and the exact energy functional open (ED's gravity being the quadratic-strain interference cross-term, not Einstein-Hilbert, could shift the latter).

On the experimental front, the 2020 underground search excluded the parameter-free Diósi-Penrose model by hunting the spontaneous radiation a *continuous* collapse must emit — and the honest finding is that **this bound does not exclude ED, and now for a computed reason, not an asserted one.** The radiation arises specifically from continuous momentum diffusion: a white-noise (Markovian Lindblad) process kicks every charge perpetually, and a randomly accelerated charge radiates. That per-

petual pumping is the price of collapsing onto a *continuous* variable (position $\hat{x}$): no physical state is a fixed point, so $d\langle p^2 \rangle / dt$ is a positive constant forever. ED collapses onto a *discrete* pointer basis instead. The corpus derives (FORCED from P11; arcs/arc-Q/lindblad_extension.md) that ED's Lindblad jump operators are **channel projectors**, $\hat{J}_\alpha = g_\alpha \Pi_\alpha$, not position operators — so a committed charge, $\rho = \Pi_\alpha$, is an **exact fixed point** of the dissipative face ($D[\Pi_\alpha] = 0$ by projector orthogonality): zero continuous momentum diffusion, zero heating, zero radiation. This settles the objection that first looked adverse — that P11's dissipative (line-gap) face would radiate: it does not, because its jump operators are bounded projectors whose fixed points are exactly the committed states, so the collapse *relaxes and halts* rather than pumping. The same fixed-point argument disposes of the more robust spontaneous-*heating* bound. Two honest limits remain. The immunity is **not a distinctive observable** — standard quantum mechanics predicts the same null; what ED adds is structural, that it is a *genuine* collapse theory (it reduces superpositions at $\tau \propto \hbar/E_G$) which nonetheless sits on the non-radiating side, unlike CSL and the noise-implemented DP, and does so because it collapses onto a discrete finite-grain basis (the same grain that regulates E_G). One **load-bearing** question had to be answered for the fixed-point argument to hold at all: an ordinary bound charge ($\sim \text{\AA}$ -nm) must sit in a *single* channel eigenstate; if ED's channel basis were fine toward the substrate scale ℓ_{ED} , ordinary matter would span many channels, the continuum limit would return CSL/DP diffusion, and ED would be *excluded*. This is now **resolved grounded-in-kind** (companion StateReduction_ChannelGranularity.md) on three standing corpus results: (1) ℓ_{ED} -fine substrate dynamics is *coherent* Schrödinger evolution (emergent unitarity), not a position measurement — the grain builds the wavefunction, it does not localize the charge; (2) the decoherence basis is *environment-selected einselection*, i.e. the coarse robust basis, and for bound charges the stationary *energy eigenbasis* (no radiation); (3) commitment individuates only *distinguishable* channels (§4), and distinguishability is *operational* (A1), with a coarse floor $\gg \ell_{ED}$. So the pointer basis is coarse, bound matter is single-channel, and the continuum limit is not reached (it would contradict emergent unitarity). CSL/DP radiate because they couple to *absolute position*, which does not commute with atomic binding; ED couples to the einselected channel the charge already occupies.

So the gap is **narrowed, and the 2020 risk retired** — immunity ED shares with textbook QM (its distinction structural, not observational), not a distinctive win. The remaining open pieces are constructive: the exact *coefficient*, the exact *energy functional* (quadratic-strain, not Einstein–Hilbert), and — now downgraded from load-bearing to a write-up task — the explicit channel-granularity theorem and its exact floor.

Alongside this stands the Report §6 caveat: the einselection-arrow = khronon-arrow identification is a structural synthesis, not forced by a deeper theorem, and it carries no *delivered* prediction — the scaling argument above is a *candidate*, not a delivered number. The honest ledger: a deep reframe, a scaling argument that reproduces the rivals' rate and improves on their framing, and an experimental risk retired to the same null status as standard QM — with the collapse coefficient, the energy functional, and the localized-charge diffusion all still open.

§6 — Position

Event Density is a member of the gravitational-collapse program, not a bystander to it, and it enters agreeing on the two convictions that define the program: geometry is not an independent quantized field, and collapse is a real physical event rooted in gravity. Where it parts company is on the *shape* of the claim. Diósi and Penrose say gravity causes collapse and put the scale in an energy, E_G . ED says the definiteness of outcomes and the flow of gravitational time are one primitive — the arrow of commitment, which is the khronon — so the collapse variable is time, and the “cause” is not a cause

but an identity. Read against Penrose specifically, ED supplies the well-defined preferred time whose *absence* drives his argument, and so reads less as his competitor than as the resolution of the gap he opens. And it does so while keeping collapse *local* and *forward-causal* — a genuinely stochastic reduction that needs no superdeterminism and no hidden variable (the A1 capacity-zero result plus the forward arrow). Read against both, ED currently owes what they deliver: a collapse rate, and the falsifiable confrontation that comes with it.

Diósi and Penrose are, in short, Event Density's *measurement-side* interlocutors, exactly as Berezhiani and Khoury are its *dark-matter-side* ones — the nearest published neighbors on each face of the same underlying bet, that gravity and the deep structure of the quantum world are one problem. Positioning ED honestly against them, deep reason and owed number alike, is how that bet gets taken seriously.

§7 — Falsifiers and tiers

Falsifiers of the ED position here: - **F1.** A computed ED collapse rate that neither recovers $\hbar/E_G$ nor matches any observed decoherence — i.e. ED gets the *rate* wrong once it is derived. (Currently the rate is not derived; this is a pending, not a passed, test.) - **F2.** A demonstration that the einselection arrow and the gravitational khronon *cannot* be the same object (e.g. the pointer basis and the gravitational foliation shown to require distinct, non-identifiable time structures) — collapses the unification move. - **F3.** Experimental confirmation of the *parameter-free* Diósi–Penrose $\hbar/E_G$ collapse with an energy-conservation signature that ED's forward-arrow, recording-based mechanism cannot accommodate. - **F4.** A demonstration that a local, forward-causal, stochastic collapse is impossible — that superdeterminism or an equivalent non-locality is *forced* for any gravitational collapse of a spread-out superposition — would remove ED's claimed locality advantage. - **(No F5 from the emission signature.)** ED's non-emission of spontaneous collapse radiation is *not* an ED-discriminating prediction — standard QM and every discrete-reduction reading predict the same null. The one route by which the emission bound *could* have tensioned ED — its dissipative (line-gap) face forcing continuous momentum diffusion on localized charges — is now closed: the corpus's FORCED jump-operator identification ($\hat{J}_\alpha = g_\alpha \hat{\Pi}_\alpha$, channel projectors) makes committed states exact fixed points, so no such diffusion exists. The one place the account could still have been tensioned — if ED's channel basis were fine toward ℓ_{ED} , ordinary matter would span many channels and ED would re-enter the excluded CSL/DP class — is now closed grounded-in-kind (companion StateReduction_ChannelGranularity.md): the pointer basis is the einselected, environment-set, distinguishability-gated *coarse* basis, and an ℓ_{ED} -fine measuring basis would contradict ED's own emergent unitarity. What *would* still tension the account is a demonstration that ED's pointer basis is ℓ_{ED} -fine after all (against the three pillars) — the residual is the explicit theorem and the exact floor, not the qualitative conclusion.

Tiers: - The arrow selects the pointer basis (einselection primitive) — **reconstruction** (Gleason/Piron–Solèr; Solèr rigor residual). Report §4. - The arrow is gravity's time; measurement-definiteness and gravitational-time are one primitive — **structural synthesis, unconfirmed**; not forced by a deeper theorem; no delivered prediction. Report §6. - Geometry emergent ($g \sim 1/b$), with both matter and geometry emergent from the substrate (empty space still has geometry; the metric carries its own dynamics) — **standing** (curvature-emergence, static half closed). - Locality without superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment) — **grounded** (A1 measured; the forward arrow is P11). - The $\tau \propto \hbar/E_G$ *scaling* from ED's own primitives (khronon-clock difference + einselection) — **grounded-in-kind scaling argument** (E_G identified with Penrose's answer, not computed; coefficient and energy functional open). Companion note. - Not excluded by the 2020 spontaneous-radiation bound — **risk retired, residual resolved from the corpus** (ED's Lindblad jump operators are discrete channel projectors $\hat{J}_\alpha = g_\alpha \hat{\Pi}_\alpha$; committed states are exact fixed

points $D[\Pi_\alpha]=0$; no continuous diffusion of a localized charge; heating bound handled by the same argument). Tier: **structural result** (FORCED identification + exact algebra); its load-bearing residual (channel basis coarser than ℓ_{ED}) is **resolved grounded-in-kind** on three standing corpus pillars (companion), residual = the explicit theorem + exact floor. *Not* a distinctive observable (standard QM predicts the same null); the distinction is structural.

*Positioning note, EDG physics-papers/state-reduction/ (working record in event-density theory/State_Reduction/). Positions ED's arrow-based state reduction (Report §4, §6) against the gravitational-collapse program: Diósi (unquantized-gravity noise, $\tau \approx \hbar/E_G$) and Penrose (ill-defined time of superposed spacetimes $\rightarrow E_G \rightarrow \tau \approx \hbar/E_G$). ED's distinctive claim: measurement-definiteness and gravity's time are one primitive (the arrow = the khronon), so the collapse variable is time, not energy; ED supplies Penrose's missing preferred time and gets locality without superdeterminism. The gap is now narrowed, not closed: the companion note gives a scaling argument reproducing $\tau \propto \hbar/E_G$ from ED's primitives (coefficient and energy functional open), and shows the 2020 spontaneous-radiation bound (which targets continuous-position-noise collapse) does not exclude ED — computed from the corpus's FORCED jump-operator identification ($\hat{J}_\alpha = g_\alpha \Pi_\alpha$, channel projectors): committed states are exact fixed points, so no continuous diffusion of a localized charge. That immunity is shared with standard QM (the distinction structural, not observational); its load-bearing prerequisite — ED's channel basis being coarser than ℓ_{ED} — is **resolved grounded-in-kind** (companion StateReduction_ChannelGranularity.md: emergent unitarity, environment-selected einselection, operational distinguishability). A candidate MOND enhancement of E_G is closed by ED's own External Field Effect (Earth's field Newtonizes it $\rightarrow ED \approx$ Penrose in the lab). Remaining open pieces are constructive: the exact coefficient/energy functional, and the explicit channel-granularity theorem + floor.*